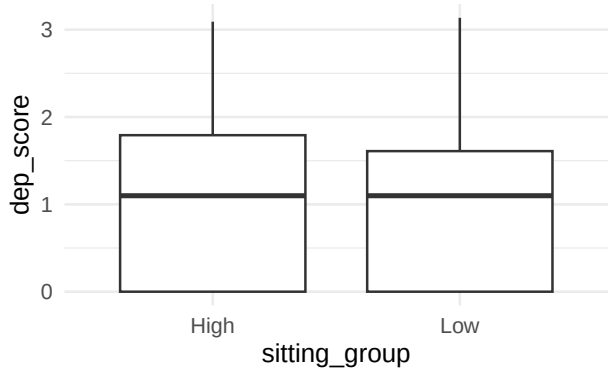
Day 02: Simple Linear Regression (SLR)

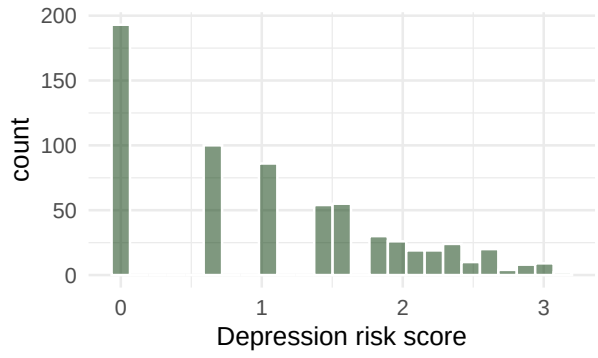
Last time

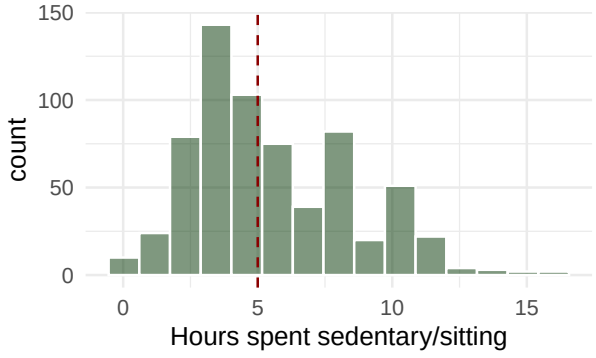
The *National Health and Nutrition Examination Survey* (NHANES) is an ongoing CDC survey that collects health and behavioral data from a nationally representative sample of the U.S. population. participants aged 18-25 were classified into two groups: those who reported **more** sitting time than average ("High"), and those who reported **less** ("Low"). Their depression symptoms were measured with the PHQ-9, a standard 9-item screening scale (0-27, higher = more symptoms). This was translated into a **depression risk score**, with a larger number indicating a higher risk.

Last time, we used the sample statistics and our trusty intro stat skills to show that there was **no significant difference** between the two groups.



A closer look:



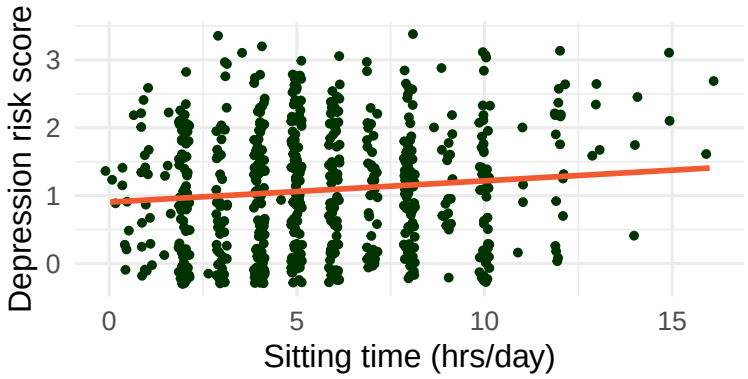


- Response variable: _____
- Predictor variable: _____

Regression, Intro Stat version

The main linear regression skills we focus on in intro stat are:

- Describe the scatterplot
- Write the equation of the regression line
- Interpret the slope and intercept
- Make a prediction



Example: describe the scatterplot

There is a _____ (strength), _____ (direction), _____ (form) association between sitting time and depression risk score.

```
lm(dep_score ~ sedentary_hours, data = nhanes)
```

```
    (Intercept) sedentary_hours
     0.90426173      0.03140912
```

Example: Write out the regression equation

Example: Interpreting the slope:

> **Correlation vs Causation**
>
> The slope describes an *association* – an increase in sitting time doesn't *cause* an increase in depression symptoms just because the model says so. Watch for language that implies causation from an

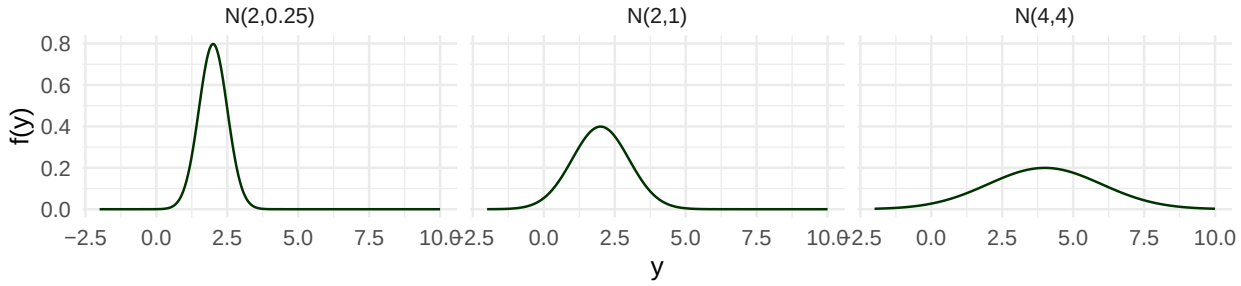
observational study.

Example: make a prediction. A participant reports sitting for 6 hours per day. The model predicts:

Regression, Stat230 Version

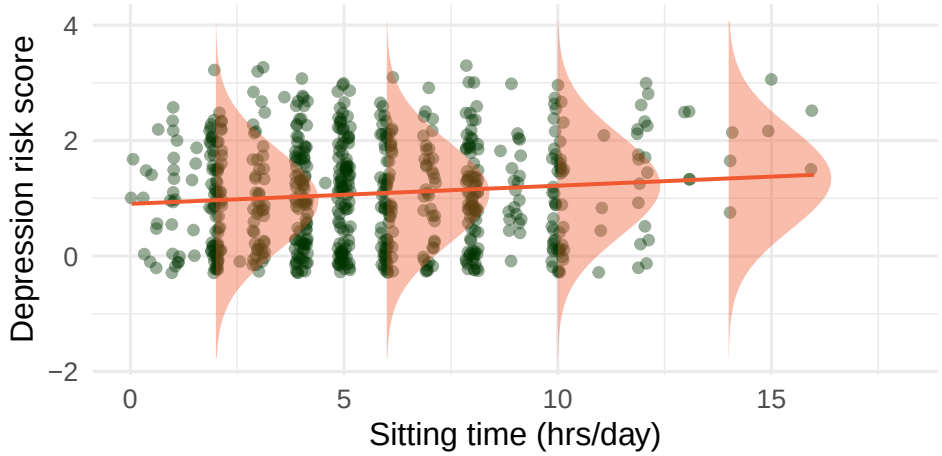
Recap: the Normal distribution

If Y is **numeric**, **unimodal**, and **symmetric**, we might use a Normal model: $Y \sim N(\mu, \sigma^2)$.



The Simple Linear Regression model

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \quad \text{where} \quad \varepsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$$



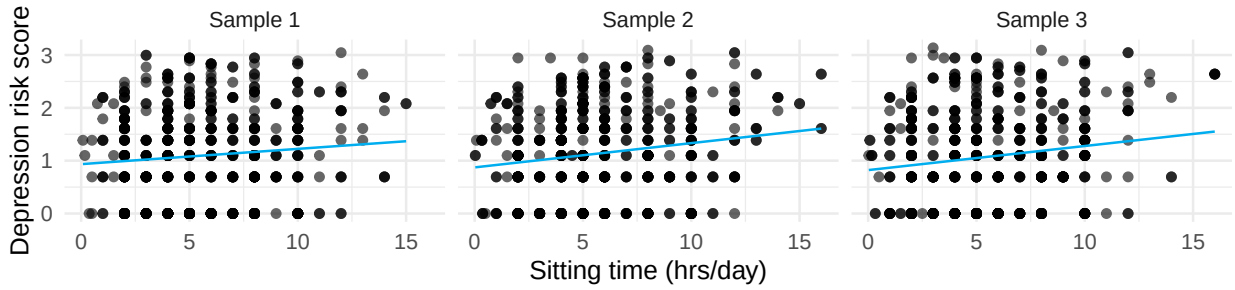
1. Linear relationship between x and y
2. Errors are independent and identically distributed (iid)
3. Errors are normally distributed
4. Errors have mean 0
5. Variance of the errors doesn't depend on x

Notation

Mean function: $E(Y \mid X) = \mu\{Y \mid X\} = \beta_0 + \beta_1 x_i$

Fitted model equation: $\hat{Y}_i = \hat{\mu}\{Y \mid X\} = \hat{\beta}_0 + \hat{\beta}_1 x_i$

Sampling Variability (preview for next class!)



R Activity

Open the SLR in R activity on Moodle and complete it in your groups. Make note of any questions, and choose someone responsible for reporting out your group's answers.

All of the R commands you need are in the file, but here's a reference list of some of the functions you'll learn:

Function	Purpose
`read.csv("file_or_url")`	Load a CSV data set from a file or the web
`glimpse(data)`	Get a quick overview of a data set
`mean(x)`	Calculate the mean
`sd(x)`	Calculate the standard deviation
`gf_histogram(~x, data = data, bins = n)`	Create a histogram
`gf_boxplot(~x, data = data)`	Create a boxplot
`gf_point(y ~ x, data = data)`	Create a scatterplot
`gf_lm()`	Add a SLR line to a scatterplot
`cor(x, y)`	Calculate the correlation
`lm(y ~ x, data = data)`	Fit a SLR model
`summary(model)`	Get detailed info about a fitted model
`confint(model, level = 0.95)`	Confidence interval for SLR coefficients
`predict(model, newdata = new_data)`	Make predictions using a fitted model